Pocket Pico HF Stuck Transmit Fix

Oct 16, 2026 · @Gene

The HF Pocket Pico could freeze with PTT stuck on when it saved its memory to flash during a transmission; the fixed firmware no longer does, and is offered to the project as pull request 11.

What happened

After running overnight on 14.105 MHz with an IC-718, the Pocket Pico locked up in transmit right after station KF7PSM-15 connected to its PBBS.

- The console stopped answering completely; pressing Enter did nothing.
- Only the PTT LED stayed lit. The radio had to be switched off to stop transmitting.
- The last console line was `TNCEMU:Saving Ram Data to Flash`. The `slot N` that normally follows never appeared.

Cause

The TNC tried to save its memory to flash while it was transmitting, and the chip froze.

1. The emulated TNC saves its memory to flash whenever the PBBS mailbox changes (or TXDELAY is changed). The connect from KF7PSM-15 triggered a save.
2. While flash is being erased, the Pico cannot read it. The transmit audio is built from tone tables that were stored in flash and streamed out automatically (DMA). A save during a transmission made that stream read flash mid-erase, which froze the chip with the PTT pin left on.
3. The firmware switched the watchdog off during saves, so nothing restarted the frozen Pico.

This bug was already in the original 1200 baud code. At 300 baud each transmission lasts four times longer, so a save now often lands during one.

The fix

Three changes, in `tnc.c` and `wave_table.h`, each guarding against it from a different side:

Change	Effect
A flash save waits until the TNC is not transmitting	Saves can no longer overlap a transmission
Tone tables moved from flash into RAM	Transmit never reads flash, even if a save did overlap
Watchdog stays on during saves (8 second limit) instead of being switched off	If anything ever freezes, the Pico restarts itself within seconds and PTT drops

Receive, transmit audio and all TNC settings are otherwise unchanged.

Test and getting it

The fix passed the test that recreates the freeze: no lock-up in three tries.

With the radio off, a long line was sent in `conv` mode, and `txd 56` was typed while the PTT LED was still lit (changing TXDELAY makes the TNC save to flash). Each time the transmission finished, the save completed with `slot N`, and the TNC kept running.

The fixed source and a ready-to-flash `pico_tnc.uf2` are on the `freeze-fix` branch of github.com/tremors834/pico_tnc. They're offered to the main project as pull request 11. To install, hold BOOTSEL while plugging in the Pico, then copy the `.uf2` file onto the RPI-RP2 drive that appears.

73, Gene N7WDZ